Practice Set

A. Arithmetic Operations

1. Calculate the sum of the first n natural numbers. Take n as input.
2. Take two numbers as input and perform addition, subtraction, and multiplication.
3. Take two numbers a and b and calculate:
   - Integer division of a by b
   - Floating-point division of a by b
   - Remainder of a divided by b
4. Take two numbers a and b and calculate a^b.
5. Find the difference between the square of the sum and the sum of the squares of the first n natural numbers.
6. Take three single-digit numbers x, y, z and print them as a single number xyz using arithmetic operations (no string concatenation).

B. Geometry & Measurement

1. Calculate the area and circumference of a circle. Take the radius as input.
2. Calculate the area of a triangle. Take base and height as input.
3. Take four integers x1, y1, x2, y2 and calculate the Euclidean distance between the points (x1, y1) and (x2, y2). Round the result to two decimal places.

C. Temperature Conversion

1. Convert a temperature from Celsius to Fahrenheit. Take the Celsius temperature as input.
2. Convert a temperature from Fahrenheit to Celsius. Take the Fahrenheit temperature as input.

D. Quadratic Equations

1. Compute the roots of a quadratic equation $ax^2+bx+c=0$, assuming the roots are real and distinct. Take a, b, c as input.

E. Number Manipulations

1. Take a number x and output x+xx+xxx+xxxx.
2. Calculate the cube of a number. The number should be user input.
3. Swap two numbers x and y and print the swapped values.

4. Take a three-digit number and print each digit on a separate line, starting from the unit's place.

F. Conditional & Logical Checks

1. Check whether a character is an alphabet (a–z or A–Z). Print 1 if it is, otherwise print 0.
2. Check whether a year is a leap year. Print 1 if it is, otherwise print 0.

G. Additional Questions

1. Write a C program that takes four integers as input from the user and prints the second largest number among them.

Conditions/Restrictions:

I. You are not allowed to use loops (`for`, `while`, `do-while`).
II. You are not allowed to use arrays or any data structures to store the numbers.
III. You must solve the problem using only nested `if-else` statements to compare the numbers.

2. Write a program that takes time input in 24-hour format as two integers:

- hh (hours, `0–23`)
- `mm` (minutes, `0–59`)

Tasks:

- If hh or `mm` are outside their valid ranges, print `"Invalid Time"`.
- Otherwise, classify the time:
  - Morning: `05:00–11:59`
  - Afternoon: `12:00–16:59`
  - Evening: `17:00–20:59`
  - Night: `21:00–23:59` and `00:00–04:59`

3. Write a program that takes three integers as input, representing day, month, and year. Your task is to determine whether these values form a valid calendar date.

Requirements:

I. Validate that the month is between 1 and 12.
II. Validate that the day is within the correct range for the given month:

    A. Months with 31 days → January, March, May, July, August, October, December.
    B. Months with 30 days → April, June, September, November.
    C. February → 28 days in a common year, 29 days in a leap year.
III. A leap year is defined as:
    A. divisible by 400, OR
    B. divisible by 4 but not by 100.
IV. If the date is valid, print "Valid Date". Otherwise, print "Invalid Date".

Example Runs:

- Input: 29 2 2024 → Output: Valid Date (2024 is a leap year)
- Input: 29 2 2023 → Output: Invalid Date (2023 is not a leap year)
- Input: 31 4 2022 → Output: Invalid Date (April has only 30 days)